

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

# Chapter 8: Logarithms

Every formula and result of the chapter on one sheet

|           |                                   |                                             |
|-----------|-----------------------------------|---------------------------------------------|
| <b>1</b>  | <b>Definition</b>                 | $a^x = N \iff \log_a N = x$                 |
| <b>2</b>  | <b>Log of 1</b>                   | $\log_a 1 = 0$                              |
| <b>3</b>  | <b>Log of the base</b>            | $\log_a a = 1$                              |
| <b>4</b>  | <b>Log of a power of the base</b> | $\log_a a^n = n$                            |
| <b>5</b>  | <b>Inverse</b>                    | $a^{\log_a N} = N$                          |
| <b>6</b>  | <b>Product law</b>                | $\log_a (mn) = \log_a m + \log_a n$         |
| <b>7</b>  | <b>Quotient law</b>               | $\log_a \frac{m}{n} = \log_a m - \log_a n$  |
| <b>8</b>  | <b>Power law</b>                  | $\log_a m^n = n \log_a m$                   |
| <b>9</b>  | <b>Root</b>                       | $\log_a \sqrt[n]{m} = \frac{1}{n} \log_a m$ |
| <b>10</b> | <b>Equal logs</b>                 | $\log_a m = \log_a n$ gives $m = n$         |
| <b>11</b> | <b>Common logs</b>                | $\log 10 = 1, \log 100 = 2, \log 0.1 = -1$  |
| <b>12</b> | <b>From log 2</b>                 | $\log 5 = 1 - \log 2$                       |